

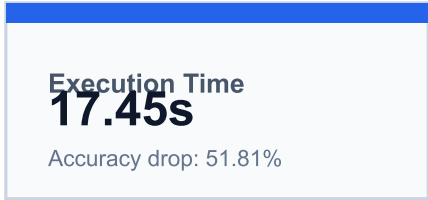
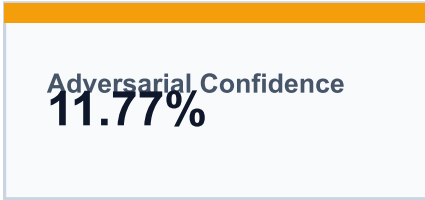
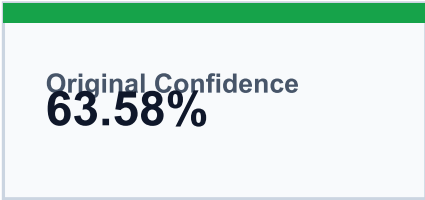
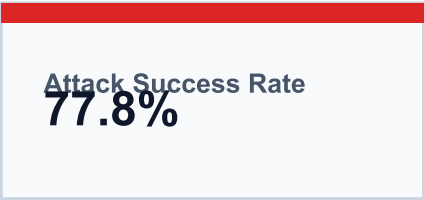
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: facebook/convnext-tiny-224

Attack Type: FGSM

Images Evaluated: 9



Threat Level Assessment
HIGH RISK

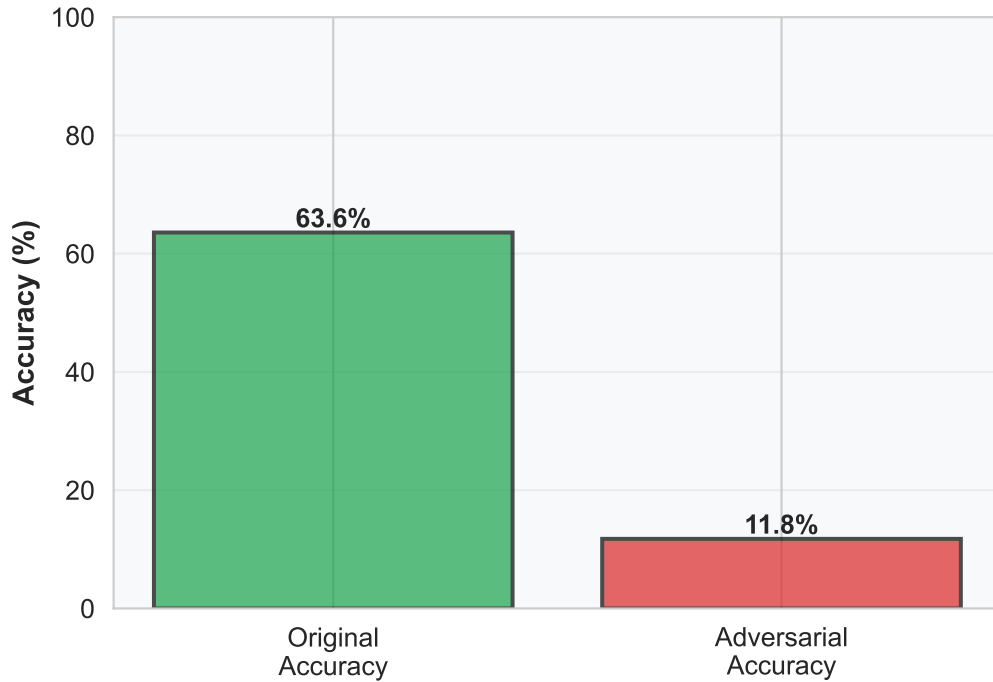
The FGSM attack altered model predictions on 77.8% of the evaluated images. Average confidence shifted from 63.58% to 11.77%, indicating the current robustness posture is high risk.

Assessment Narrative

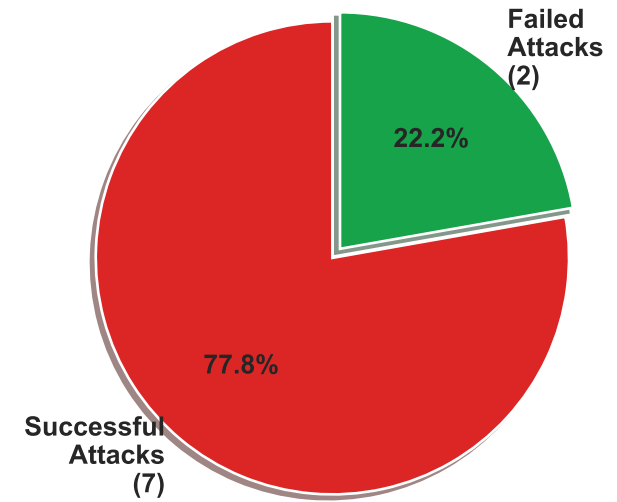
Successfully executed FGSM attack on model facebook/convnext-tiny-224 using 9 test images. Attack success rate: 77.8%. Average original accuracy: 63.58%, Average adversarial accuracy: 11.77%. The attack successfully fooled the model in 7 out of 9 cases.

Detailed Analysis

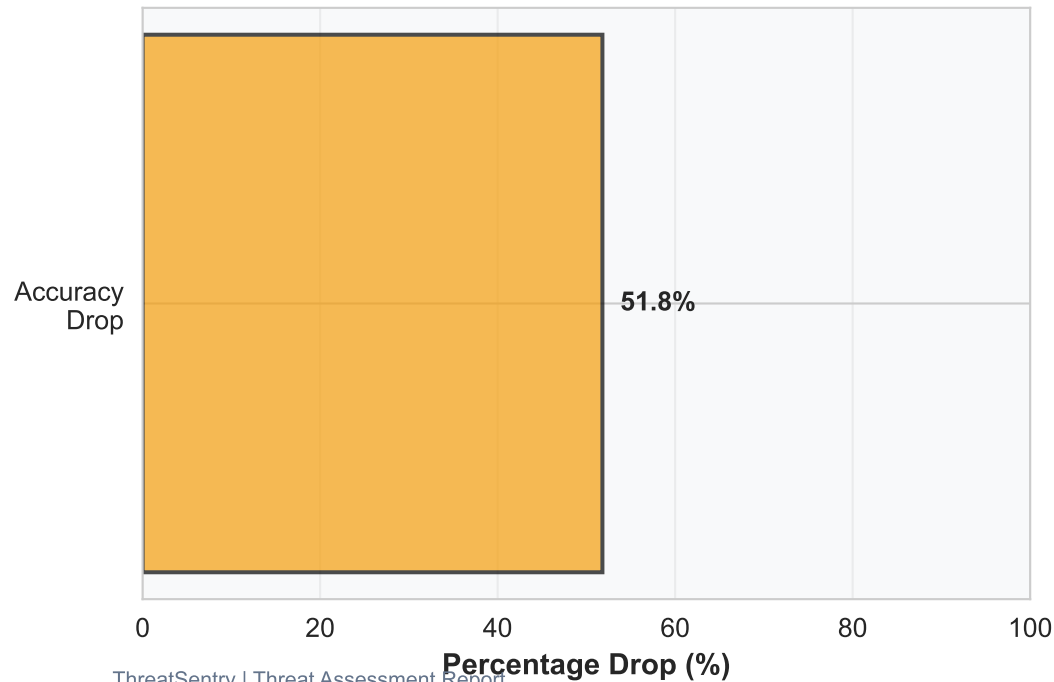
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

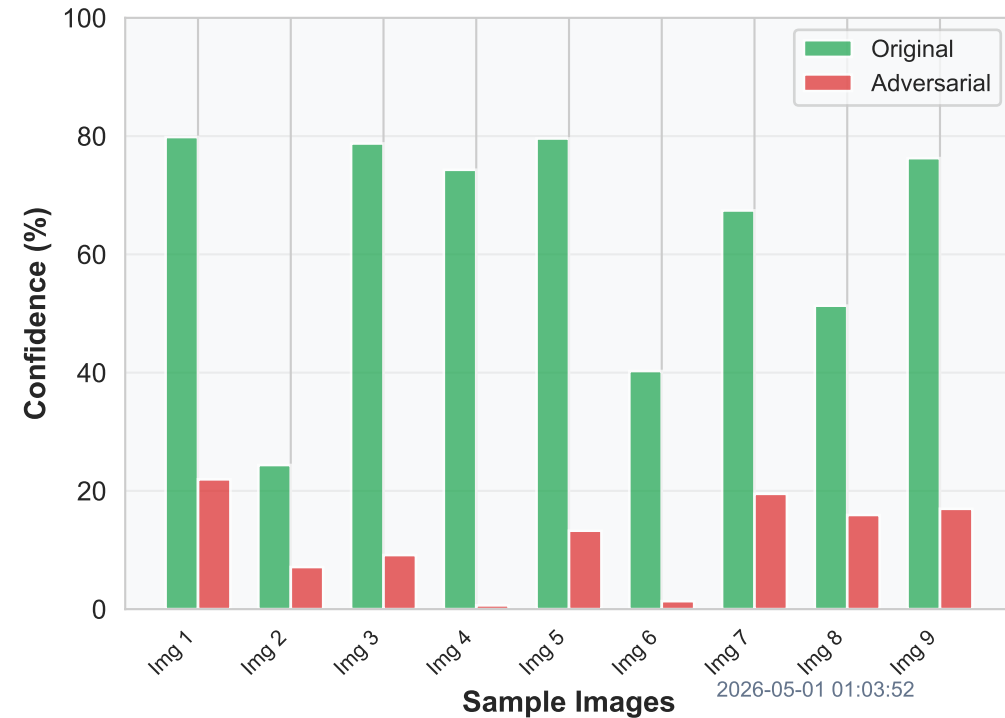


Image Evidence - Original vs Adversarial

Images 1-2 of 9

Original: dhole, Cuon alpinus
(79.6%)



Adversarial: dhole, Cuon alpinus
(13.3%)
ATTACK BLOCKED



Original: coyote, prairie wolf, brush wolf, Canis latrans
(51.3%)



Adversarial: dhole, Cuon alpinus
(15.9%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 3-4 of 9

Original: Indian elephant, *Elephas maximus*
(67.4%)



Adversarial: tusker
(19.5%)
ATTACK SUCCESSFUL



Original: ibex, *Capra ibex*
(76.3%)



Adversarial: gazelle
(17.0%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 5-6 of 9

Original: tiger, *Panthera tigris*
(79.8%)



Adversarial: tiger, *Panthera tigris*
(22.0%)
ATTACK BLOCKED



Original: gibbon, *Hylobates lar*
(78.8%)



Adversarial: howler monkey, howler
(9.2%)
ATTACK SUCCESSFUL

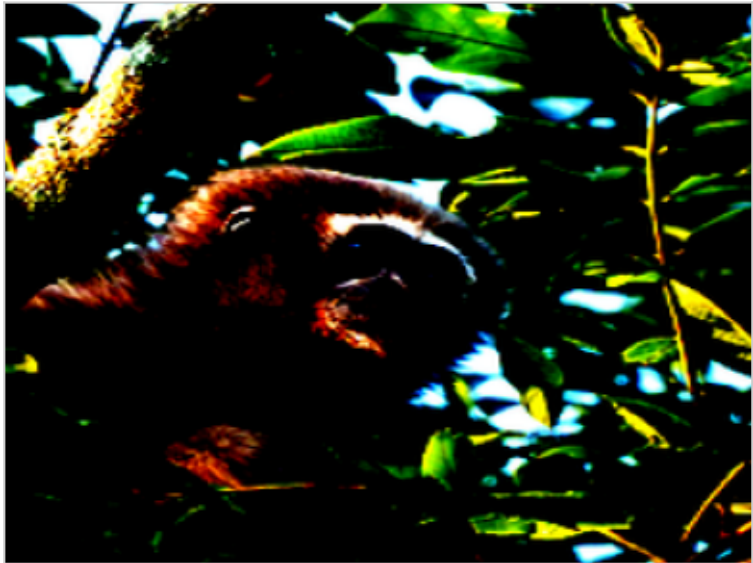


Image Evidence - Original vs Adversarial

Images 7-8 of 9

Original: capuchin, ringtail, Cebus capucinus
(40.3%)



Adversarial: bee eater
(1.4%)
ATTACK SUCCESSFUL



Original: ox
(24.4%)



Adversarial: hippopotamus, hippo, river horse, Hippopotamus amphibius
(7.1%)
ATTACK SUCCESSFUL

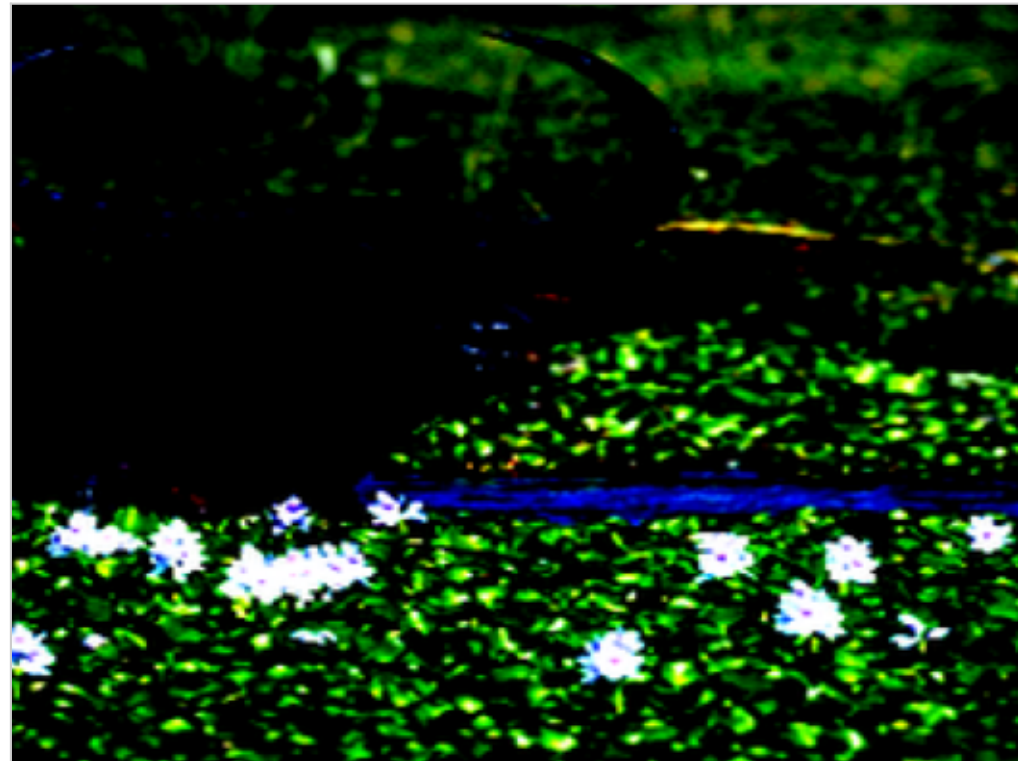


Image Evidence - Original vs Adversarial

Images 9-9 of 9

Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis
(74.3%)

Adversarial: impala, Aepyceros melampus
(0.6%)
ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model facebook/convnext-tiny-224 using 9 test images. Attack success rate: 77.8%. Average original accuracy: 63.58%, Average adversarial accuracy: 11.77%. The attack successfully fooled the model in 7 out of 9 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques